

$$1. \quad X = \begin{pmatrix} 1 & -1 \\ 1 & -1 \\ 1 & 1 \\ 1 & 2 \end{pmatrix}$$

• X^+ :

$$X^T = \begin{pmatrix} 1 & 1 & 1 & 1 \\ -1 & -1 & 1 & 2 \end{pmatrix}$$

$$X^T X = \begin{pmatrix} 4 & -1 - 1 + 1 + 2 \\ 1 & 1 + 1 + 1 + 4 \end{pmatrix} = \begin{pmatrix} 4 & 1 \\ 1 & 7 \end{pmatrix}$$

$$(X^T X)^{-1} = \frac{1}{4 \cdot 7 - 1 \cdot 1} \begin{pmatrix} 7 & -1 \\ -1 & 4 \end{pmatrix} = \begin{pmatrix} \frac{7}{27} & -\frac{1}{27} \\ -\frac{1}{27} & \frac{4}{27} \end{pmatrix}$$

$$X^+ = \begin{pmatrix} \frac{7}{27} & -\frac{1}{27} \\ -\frac{1}{27} & \frac{4}{27} \end{pmatrix} \begin{pmatrix} 1 & 1 & 1 & 1 \\ -1 & -1 & 1 & 2 \end{pmatrix} = \begin{pmatrix} \frac{7}{27} + \frac{1}{27} & \frac{8}{27} & \frac{6}{27} & \frac{5}{27} \\ -\frac{1}{27} - \frac{4}{27} & \frac{-5}{27} & \frac{3}{27} & \frac{7}{27} \end{pmatrix} =$$

$$\begin{pmatrix} \frac{8}{27} & \frac{8}{27} & \frac{6}{27} & \frac{5}{27} \\ \frac{-5}{27} & \frac{-5}{27} & \frac{3}{27} & \frac{7}{27} \end{pmatrix}$$

$$\Theta^* = \begin{pmatrix} \frac{8}{27} & \frac{8}{27} & \frac{6}{27} & \frac{5}{27} \\ \frac{-5}{27} & \frac{-5}{27} & \frac{3}{27} & \frac{7}{27} \end{pmatrix} \begin{pmatrix} -1 \\ 1 \\ 2 \\ 3 \end{pmatrix} = \begin{pmatrix} -\frac{8}{27} + \frac{8}{27} + \frac{12}{27} + \frac{15}{27} \\ \frac{5}{27} + \frac{21}{27} \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

$$\begin{aligned} J(\Theta^*) &= \sum_{i=1}^4 \left(\left(\begin{pmatrix} 1 \\ 1 \end{pmatrix}^T \begin{pmatrix} 1 \\ x^{(i)} \end{pmatrix} - y^{(i)} \right)^2 = \left(\begin{pmatrix} 1 \\ 1 \end{pmatrix}^T \begin{pmatrix} 1 \\ -1 \end{pmatrix} + 1 \right)^2 + \\ &\left(\begin{pmatrix} 1 \\ 1 \end{pmatrix}^T \begin{pmatrix} 1 \\ -1 \end{pmatrix} - 1 \right)^2 + \left(\begin{pmatrix} 1 \\ 1 \end{pmatrix}^T \begin{pmatrix} 1 \\ 1 \end{pmatrix} - 2 \right)^2 + \left(\begin{pmatrix} 1 \\ 1 \end{pmatrix}^T \begin{pmatrix} 1 \\ 2 \end{pmatrix} - 3 \right)^2 = \\ &(1 \cdot 1 + 1 \cdot (-1) + 1)^2 + (-1)^2 + (1 \cdot 1 + 1 \cdot 1 - 2)^2 + (1 \cdot 1 + 1 \cdot 2 - 3)^2 = \\ &1 + 1 = 2 \end{aligned}$$

$$2. J(\theta) = \sum_{i=1}^4 \left(\begin{pmatrix} \theta_0 \\ \theta_1 \end{pmatrix}^T \begin{pmatrix} 1 \\ x^{(i)} \end{pmatrix} - y^{(i)} \right)^2 = \left(\begin{pmatrix} \theta_0 \\ \theta_1 \end{pmatrix}^T \begin{pmatrix} 1 \\ -1 \end{pmatrix} - 1 \right)^2 +$$

$$\left(\begin{pmatrix} \theta_0 \\ \theta_1 \end{pmatrix}^T \begin{pmatrix} 1 \\ -1 \end{pmatrix} - 1 \right)^2 + \left(\begin{pmatrix} \theta_0 \\ \theta_1 \end{pmatrix}^T \begin{pmatrix} 1 \\ 1 \end{pmatrix} - 2 \right)^2 + \left(\begin{pmatrix} \theta_0 \\ \theta_1 \end{pmatrix}^T \begin{pmatrix} 1 \\ 2 \end{pmatrix} - 3 \right)^2 =$$

$$\begin{aligned} & (\theta_0 - (\theta_1 + 1))^2 + (\theta_0 - (\theta_1 - 1))^2 + (\theta_0 + \theta_1 - 2)^2 + (\theta_0 + (2\theta_1 - 3))^2 = \\ & \theta_0^2 - 2\theta_0(\theta_1 + 1) + (\theta_1 + 1)^2 + \theta_0^2 - 2\theta_0(\theta_1 - 1) + (\theta_1 - 1)^2 + \theta_0^2 + \\ & 2\theta_0(\theta_1 - 2) + (\theta_1 - 2)^2 + \theta_0^2 + 2\theta_0(2\theta_1 - 3) + (2\theta_1 - 3)^2 = \\ & 4\theta_0^2 - 2\cancel{\theta_0\theta_1} - 2\cancel{\theta_0} + \theta_1^2 + 2\cancel{\theta_1} + 1 - 2\theta_0\theta_1 + 2\cancel{\theta_0} + \theta_1^2 - 2\cancel{\theta_1} + 1 \\ & + 2\cancel{\theta_0\theta_1} - 4\theta_0 + \theta_1^2 - 4\theta_1 + 4 + 4\theta_0\theta_1 - 6\theta_0 + 4\theta_1^2 - 12\theta_1 + 4 = \\ & 4\theta_0^2 + 2\theta_0\theta_1 + 7\theta_1^2 - 10\theta_0 - 16\theta_1 + 15 \end{aligned}$$

• $\nabla J(\theta)$:

$$\frac{\partial J}{\partial \theta_0}(\theta) = 8\theta_0 + 2\theta_1 - 10$$

$$\frac{\partial J}{\partial \theta_1}(\theta) = 2\theta_0 + 14\theta_1 - 16$$

$$\nabla J(\theta) = \begin{pmatrix} 8\theta_0 + 2\theta_1 - 10 \\ 2\theta_0 + 14\theta_1 - 16 \end{pmatrix}$$

$$\bullet \nabla J(\theta^*) = \begin{pmatrix} 8 \cdot 1 + 2 \cdot 1 - 10 \\ 2 \cdot 1 + 14 \cdot 1 - 16 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

$$3. f_{\theta^*}(1.5) = \theta_0^* + \theta_1^* \cdot 1.5 = 1 + 1 \cdot 1.5 = 2.5$$

- $N_2(1.5) = \{1, 2\}$

$$\hat{y} = \frac{1}{2} \sum_{i: x_i \in N_2(1.5)} y_i = \frac{1}{2} (2 + 3) = 2.5$$